

RED HAT ENTERPRISE LINUX

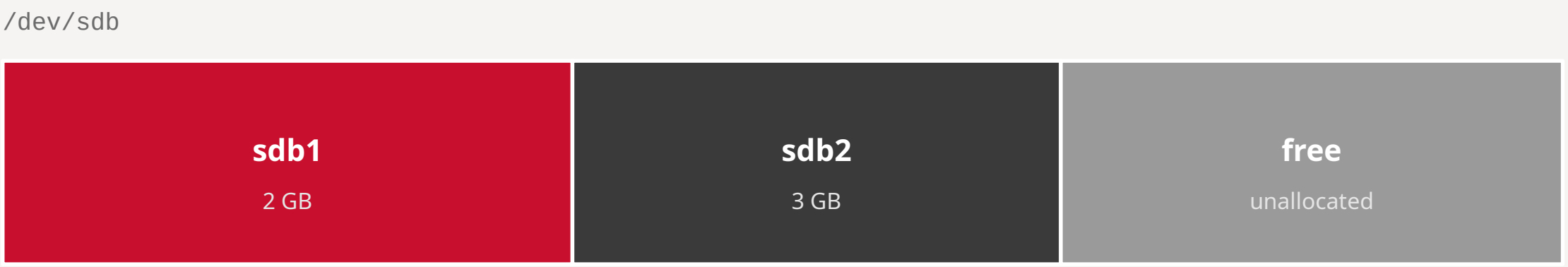
Disk Partitioning & Mounting

Partition → Format → Mount → Persist

9 SLIDES · SYSTEM ADMINISTRATION

What Is a Partition?

A logical slice of a physical disk



One physical disk can hold multiple partitions — each behaves like its own independent storage device.

MBR

up to 4 primary partitions

GPT

up to 128 partitions

The Tools You'll Use

Three commands cover the whole workflow

1

`lsblk`

See disks & partitions

2

`fdisk / parted`

Create & edit partitions

3

`mkfs`

Format with a filesystem

Create a Partition

Four keystrokes inside fdisk

```
sudo fdisk /dev/sdb
```

1

```
n
```

New partition

2

```
p
```

Primary type

3

```
+2G
```

Set size

4

```
w
```

Write & exit

Then format & verify:

```
lsblk
```

Format the Partition

Give it a filesystem before it can store files

XFS

```
mkfs.xfs /dev/sdb1
```

RHEL default

ext4

```
mkfs.ext4 /dev/sdb1
```

Widely compatible

vfat

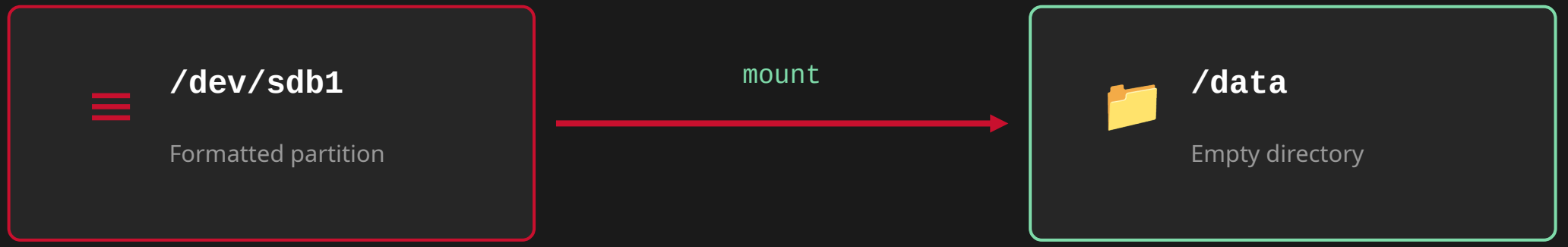
```
mkfs.vfat /dev/sdb1
```

Cross-platform

Pick one filesystem per partition — formatting erases existing data.

What Is a Mount Point?

A directory that gives the partition an address in the filesystem tree



Files written to /data now live on /dev/sdb1.

```
sudo mkdir /data
```

Mount It

Attach the partition to its mount point — active until reboot

```
sudo mount /dev/sdb1 /data
```

```
df -h
```

confirm it's mounted

```
mount | grep sdb1
```

see mount options

```
lsblk
```

view the mountpoint column

A plain mount is temporary — it disappears on reboot.

Make the Mount Persistent

Add an entry to /etc/fstab so it survives reboot

```
blkid /dev/sdb1
```

get its UUID

```
sudo nano /etc/fstab

UUID=xxxx-xxxx  /data  xfs  defaults  0 0
```

device · mountpoint · fstype · options · dump · fsck

✓ **Test before rebooting**

```
sudo mount -a
```

Re-mounts everything in fstab — catches typos immediately instead of at next boot.

No errors + df -h shows /data?

You're done — the mount survives every future reboot.

Recap

The full path from raw disk to persistent storage

